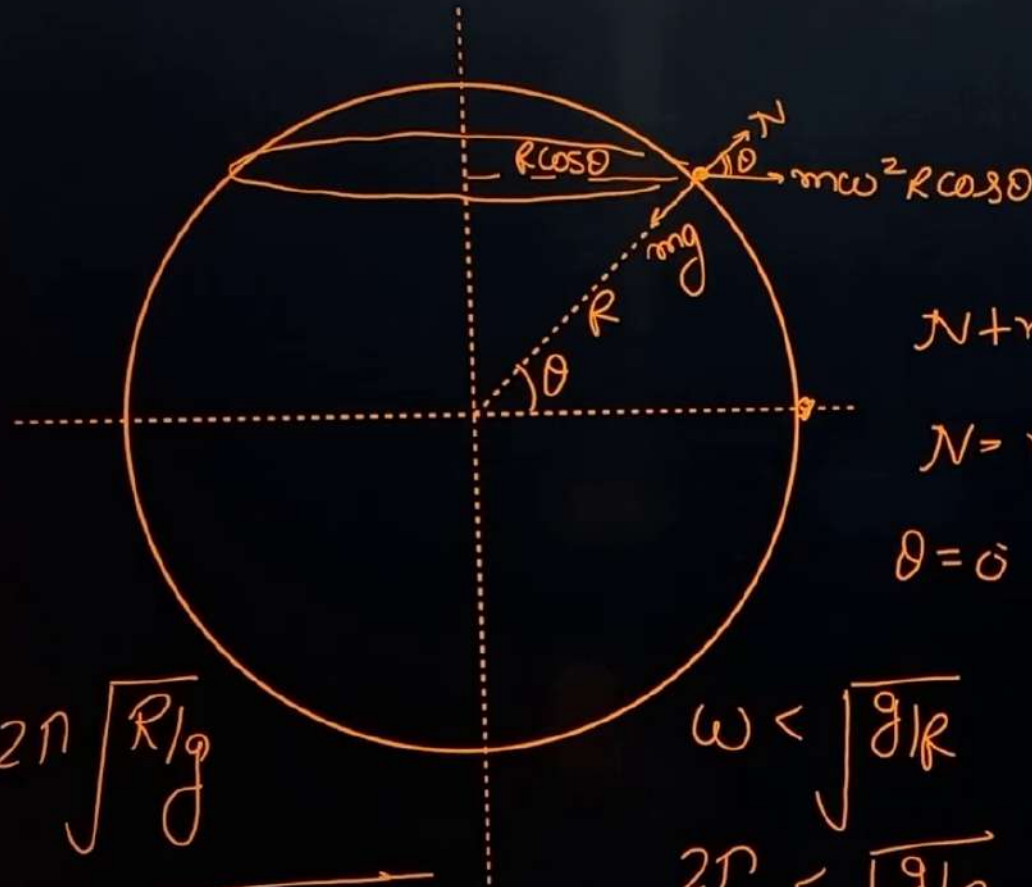


If a planet rotates too fast, rocks from its surface will start flying off its surface. If density of a homogeneous planet is ρ and material is not flying off its surface then show that its time period of

rotation must be greater than $\sqrt{\frac{3\pi}{G\rho}}$.



$$N + m\omega^2 R \cos^2 \theta = mg$$

$$N = mg - m\omega^2 R \cos^2 \theta$$

$$\theta = 0 \quad N = mg - m\omega^2 R > 0$$

$$g > \omega^2 R$$

$$T > 2\pi \sqrt{R/g}$$

$$T > 2\pi \sqrt{\frac{R \cdot R^3}{G \cdot 4\pi R^3}}$$

$$\omega < \sqrt{g/R}$$

$$\frac{2\pi}{T} < \sqrt{g/R}$$

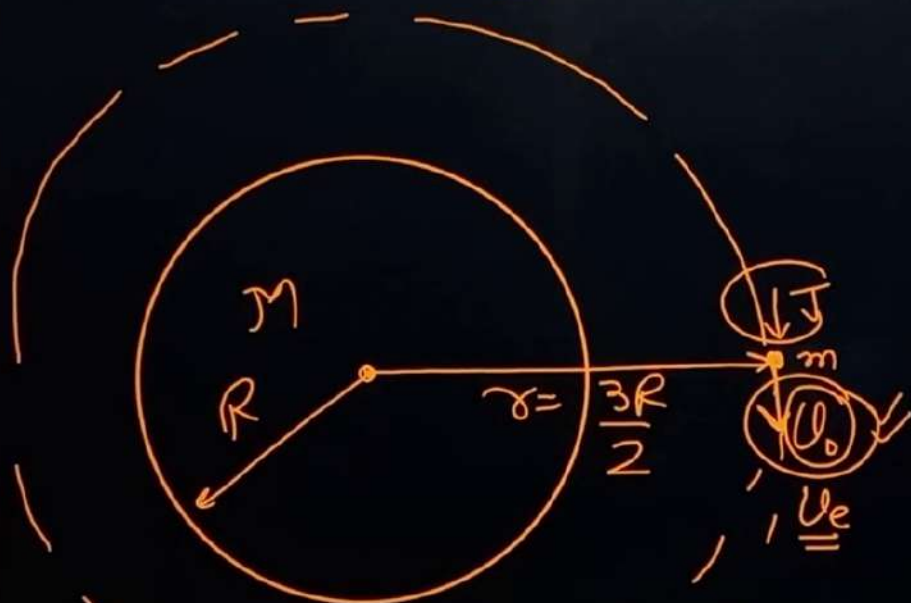
$$T > \sqrt{\frac{3\pi}{G\rho}}$$

$$U_0 = \sqrt{\frac{GM}{r}}$$

$$U_e = \sqrt{\frac{2GM}{r}}$$

$$J = mU_e - mU_0$$

$$= m(\sqrt{2}-1)U_0 = m(\sqrt{2}-1)\sqrt{\frac{GM}{3R}}$$



$$\frac{1}{2}m \frac{GM}{3R}$$

$$W.D. \text{ of Rocket} = \Delta K = K_f - K_i$$

$$= \frac{1}{2}mU_e^2 - \frac{1}{2}mU_0^2$$

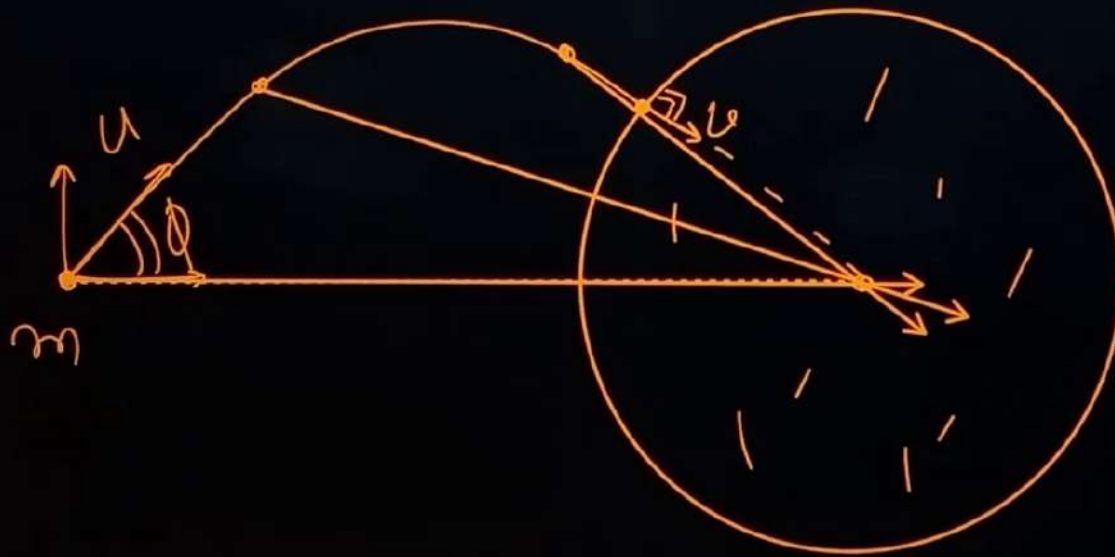
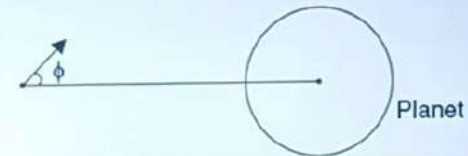
$$= \frac{1}{2}m(2U_0^2 - U_0^2) = \frac{1}{2}mU_0^2$$

$$= \frac{GMm}{3R}$$

A satellite of mass m is going around the earth in a circular orbital at a height $\frac{R}{2}$ from the surface of the earth. The satellite has lived its life and a rocket, on board, is fired to make it leave the gravity of the earth. The rocket remains active for a very small interval of time and imparts an impulse in the direction of motion of the satellite. Neglect any change in mass due to firing of the rocket.

- Find the minimum impulse imparted by the rocket to the satellite.
- Find the minimum work done by the rocket engine. Mass of the earth = M , Radius of the earth = R

A small asteroid is at a large distance from a planet and its velocity makes an angle ϕ ($\neq 0$) with line joining the asteroid to the centre of the planet. Prove that such an asteroid can never fall normally on the surface of the planet.

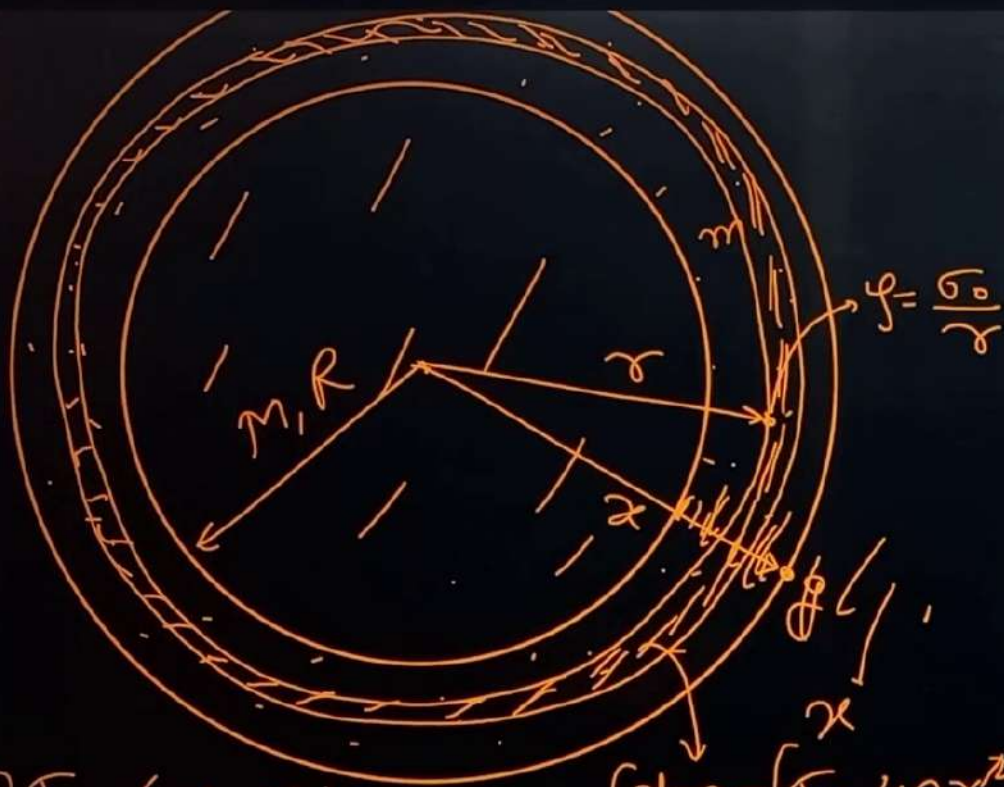


$$z=0$$

$$L_i = \text{const}$$

$$m u \sin \phi \neq 0$$

✓ Proved



$$\frac{4\pi\sigma_0}{2} (x^2 - R^2)$$

$$m = \int dm = \int \frac{\sigma_0}{r} 4\pi r^2 dr$$

$$g = \frac{2\pi G M}{R^2} = \frac{G \sigma_0}{R^2}$$

having density $\rho = \frac{\sigma_0}{r}$ where σ_0 is a constant and r is distance from the centre of the planet. It is found that acceleration due to gravity is constant throughout the atmosphere of the planet. Find σ_0 in terms of M and R .

$$g = \frac{G(M+m)}{x^2}$$

$$g = \frac{GM}{x^2} + \frac{G 2\pi\sigma_0 (x^2 - R^2)}{x^2}$$

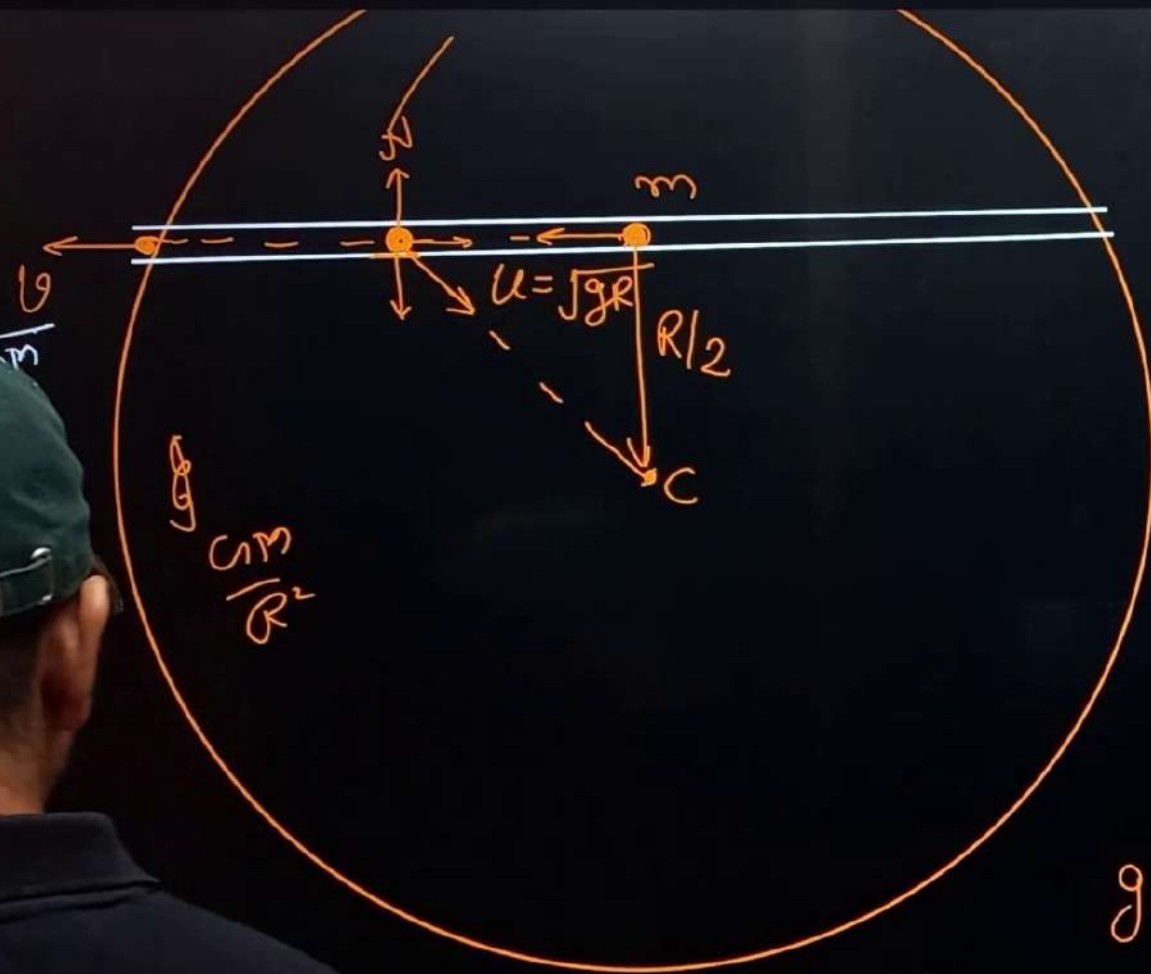
$$g = \left(\frac{GM}{x^2} \right) + 2\pi G \sigma_0 - \frac{2\pi G \sigma_0 R^2}{x^2}$$

$$GM = 2\pi G \sigma_0 R^2$$

$$\sigma_0 = \frac{M}{2\pi R^2}$$

along the tunnel from the centre of the tunnel at a speed $V_0 = \sqrt{gR}$ [g is acceleration due to gravity at the surface of the earth].

- (a) Is the angular momentum [about the centre of the earth] of the projectile conserved as it moves along the tunnel?
- (b) Calculate the maximum distance of the projectile from the centre of the earth during its course of motion.



$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2} m g R - \frac{m G M}{2 R^3} \left\{ 3 R^2 - \frac{R^2}{4} \right\}$$

$$= \frac{1}{2} m v^2 - \frac{m G M}{R}$$

$$g R - \frac{G M}{R^3} \left(3 R^2 - \frac{R^2}{4} \right) + \frac{2 G M}{R} = v^2$$

$$v^2 = \frac{G M}{R} - \frac{3 G M}{R} + \frac{G M}{4 R} + \frac{2 G M}{R} = \frac{G M}{4 R} \left\{ \right.$$

$$v = \sqrt{\frac{G M}{4 R}}$$

$$v = \sqrt{\frac{GM}{4R}}$$

$$L_i = L_f$$

$$m v R \sin 30^\circ = m v' r \sin 90^\circ$$

$$\boxed{\frac{vR}{2} = v'r} \quad \text{--- (1)}$$

$$\begin{aligned} v'^2 &= \frac{v^2 R^2}{4r^2} \\ &= \frac{GM}{4R} \cdot \frac{R^2}{4r^2} \end{aligned}$$

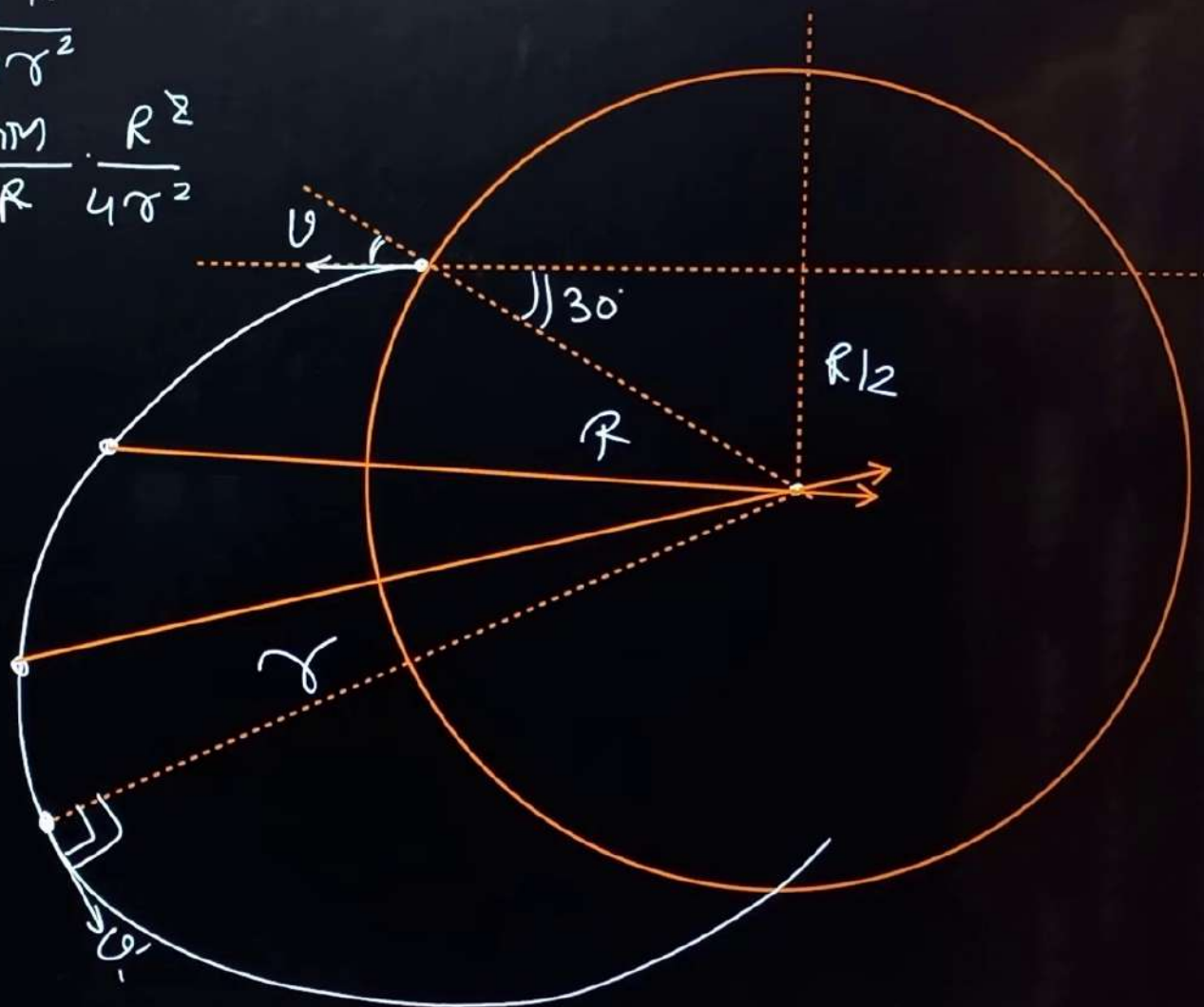
TMEC

$$\frac{1}{2} m \frac{GM}{4R} - \frac{GM}{R} m = \frac{1}{2} m \frac{GM R}{16r^2}$$

$$- \frac{mGM}{2R} = \frac{mGM R}{32r^2}$$

$$\frac{1}{8R} - \frac{1}{R} = \frac{R}{32r^2} - \frac{1}{r}$$

$$-\frac{7}{8R} = \frac{R}{32r^2} - \frac{1}{r}$$



$$-\frac{7}{8R} \times 32\gamma^2 = R - \frac{1}{\gamma} \times 32\gamma^2$$

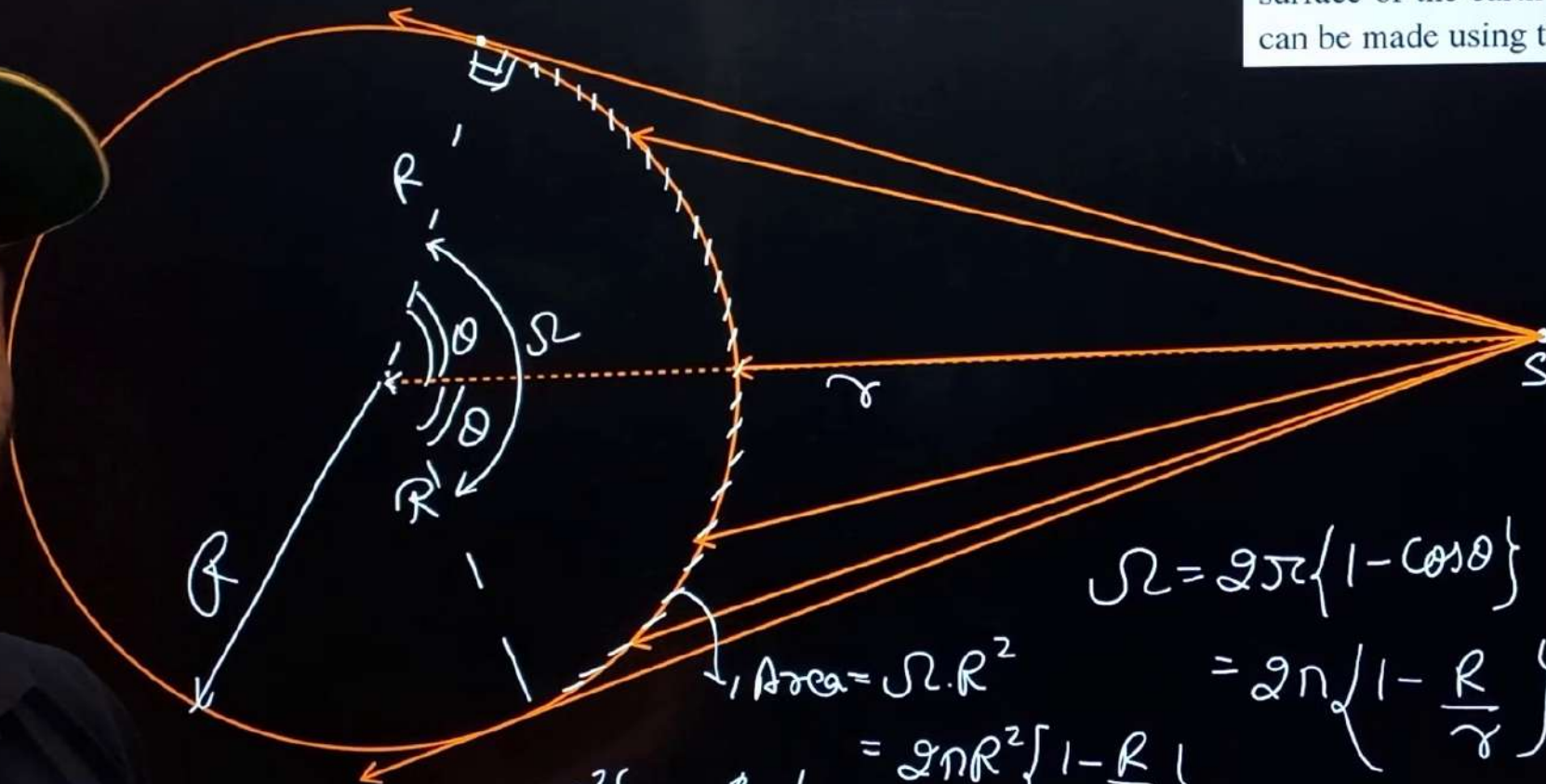
$$-28\gamma^2 = R^2 - 32\gamma R$$

$$28\gamma^2 - (32R)\gamma + R^2 = 0$$

$$\gamma = \frac{+32R + \sqrt{(32R)^2 - 4 \times 28 \cdot R^2}}{2 \times 28}$$

$$\cos \theta = \frac{R}{r}$$

A geostationary satellite is nearly at a height of $h = 6R$ from the surface of the earth where R is the radius of the earth. Calculate the area on the surface of the earth in which the communication can be made using this satellite.

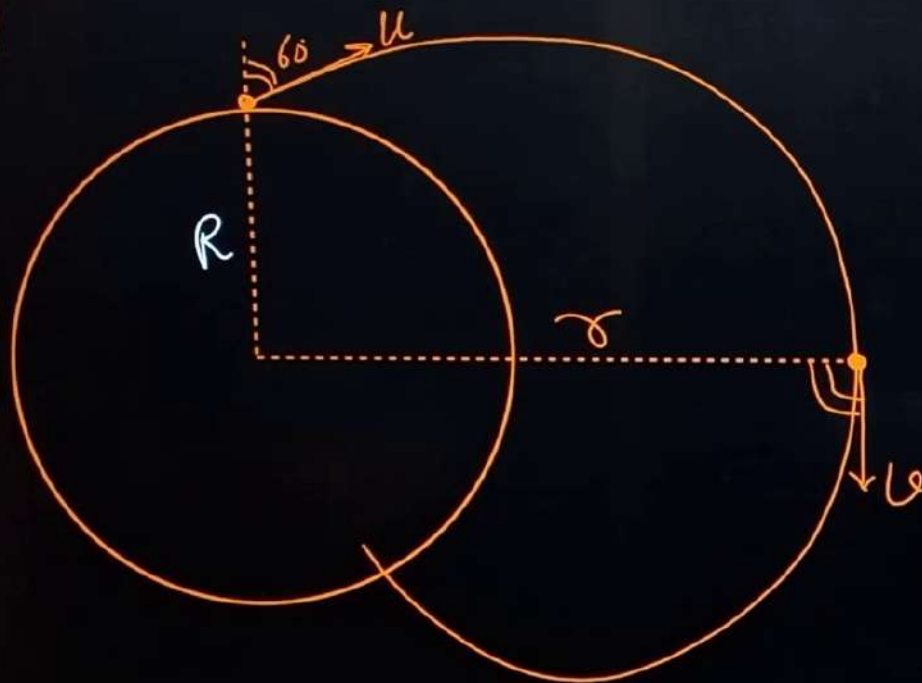


$$\begin{aligned}\Omega &= 2\pi \left\{ 1 - \cos \theta \right\} \\ &= 2\pi \left\{ 1 - \frac{R}{r} \right\}\end{aligned}$$

$$\begin{aligned}\text{Area} &= 2\pi R^2 \left\{ 1 - \frac{R}{7R} \right\} \\ &= 2\pi R^2 \left\{ 1 - \frac{1}{7} \right\} \\ &= \frac{12\pi R^2}{7}\end{aligned}$$

$$u = \frac{v_e}{2} = \frac{\sqrt{\frac{2cm}{R}}}{2}$$

$$u = \sqrt{\frac{cm}{2R}}$$



A body is projected up from the surface of the earth with a velocity half the escape velocity at an angle of 30° with the horizontal. Neglecting air resistance and earth's rotation, find

- the maximum height above the ~~earth~~'s surface to which the body will rise. (R) given
- will the body move around the ~~earth~~ as a satellite?

AMCT

$$m u R \sin 60^\circ = m U r \sin 90^\circ$$

$$\boxed{\frac{u R \sqrt{3}}{2} = U r}$$

TmEC

$$\frac{1}{2} m u^2 - \frac{GMm}{R} = \frac{1}{2} m v^2 - \frac{GMm}{r}$$

$$m \frac{GM}{2R} - \frac{2GMm}{R} = m \frac{3}{4} \frac{u^2 R^2}{r^2} - \frac{2GMm}{r}$$

$$-\frac{3}{2} \frac{GM}{R} = \frac{3}{4} \frac{R^2}{r^2} \frac{GM}{2R} - \frac{2GM}{r}$$

$$-\frac{3}{2R} = \frac{3R}{8r^2} - \frac{2}{r}$$

$$-\frac{3}{2R} \times 8r^2 = 3R - \frac{2}{r} \times 8r^2 \Rightarrow -\frac{12r^2}{R} = 3R - 16r$$

$$-12r^2 = 3R^2 - 16rR$$

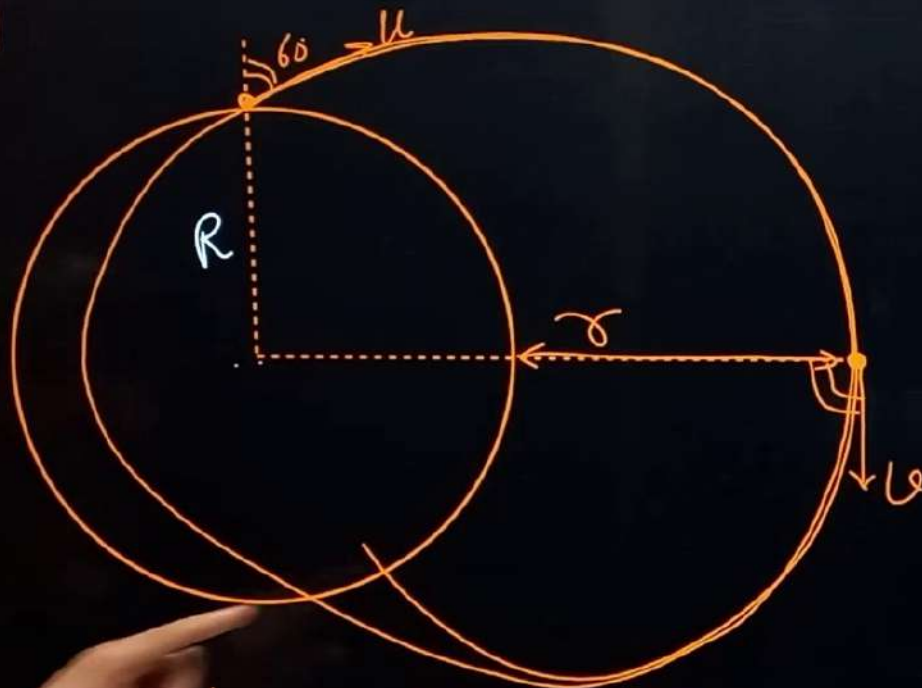
$$12r^2 - (16R)r + 3R^2 = 0$$

$$r = \frac{(4 \pm \sqrt{7})R}{6}$$

$$\boxed{r = \left(\frac{4 + \sqrt{7}}{6} \right) R}$$

$$u = \frac{v_e}{2} = \frac{\sqrt{\frac{2cm}{R}}}{2}$$

$$u = \sqrt{\frac{cm}{2R}}$$



$$h = x - R = \frac{4 + \sqrt{7}}{6} R - R$$

$$= \left(\frac{\sqrt{7} - 2}{6} \right) R$$

A body is projected up from the surface of the earth with a velocity half the escape velocity at an angle of 30° with the horizontal. Neglecting air resistance and earth's rotation, find

- the maximum height above the ~~earth's~~ surface to which the body will rise. (R) given
- will the body move around the ~~earth~~ as a satellite?

AMCT

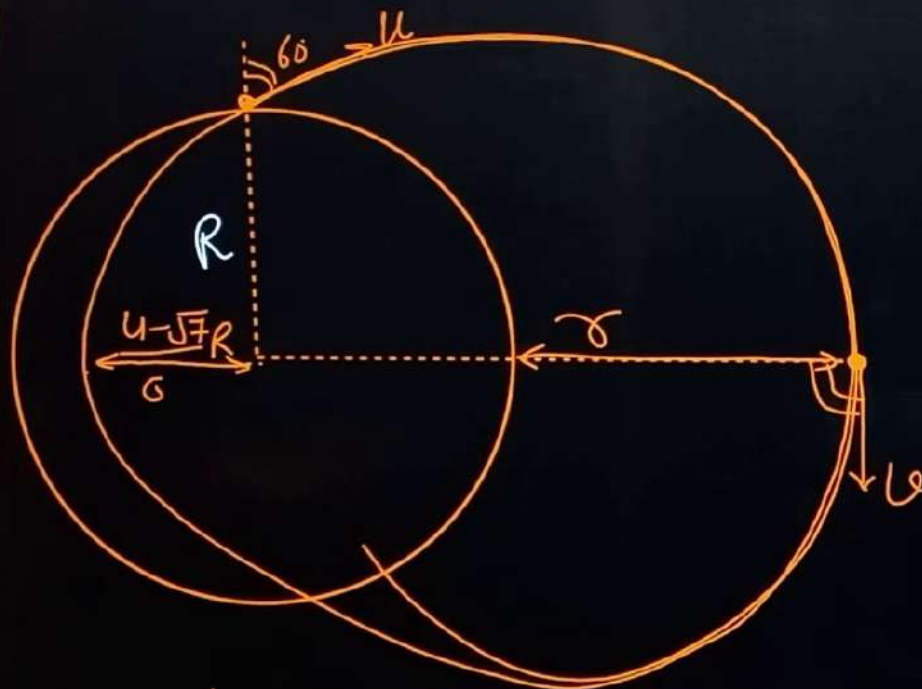
$$m u R \sin 60 = m v x \sin 90$$

$$\boxed{\frac{u R \sqrt{3}}{2} = v x}$$

$$v^2 = \frac{3}{4} \frac{u^2 R^2}{x^2}$$

$$u = \frac{v_e}{2} = \frac{\sqrt{\frac{2cm}{R}}}{2}$$

$$u = \sqrt{\frac{cm}{2R}}$$



$$h = r - R = \frac{4 + \sqrt{7} R}{6} - R$$

$$= \left(\frac{\sqrt{7} - 2}{6} \right) R$$

A body is projected up from the surface of the earth with a velocity half the escape velocity at an angle of 30° with the horizontal. Neglecting air resistance and earth's rotation, find

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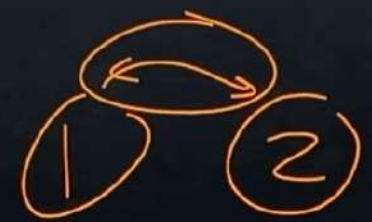
AMCT

$$m u R \sin 60 = m U r \sin 90$$

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$$U^2 = \frac{3}{4} \frac{u^2 R^2}{r^2}$$

JEE adv.



① night-sleep.

② time

③ Instruction/marking scheme

④ Roller coaster
Zhang

More No-negative

